

Tuan M. Truong

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Education

National University of Singapore

Aug 2026 – Current

PhD Student, School of Computing

- Supervisor: Prof. Shuicheng Yan

University of British Columbia

Sept 2020 – May 2024

BSc, Combined Honours in Mathematics and Computer Science

- GPA: 87.32% (4.1/4.33; A)

Honours/Awards

Bronze Medal , 59th International Mathematical Olympiad (IMO)	2018
Second Prize , Vietnamese Mathematical Olympiad (VMO)	2018
Third Prize , Vietnamese Mathematical Olympiad (VMO)	2017
UBC Trek Excellence Scholarship for Continuing Students - \$4000	2023
UBC Computer Science Scholarship - \$1400	2023
UBC Faculty of Science International Student Scholarship - \$7500	2022, 2023
First Runner-up , DENSO Factory Hacks Hackathon Challenge	2023

Publications

(*) denotes equal contribution

- **T. Truong***, C. Nguyen*, H. Nguyen*, M. Le, N. Ho, and T. Le. RepLoRA: Reparameterizing Low-rank Adaptation via the Perspective of Mixture of Experts. *International Conference on Machine Learning (ICML)*, 2025.
- **T. Truong***, Q. Tran*, N. Q. Pham, N. Ho, D. Phung, T. Le. Improving Generalization with Flat Hilbert Bayesian Inference. *International Conference on Machine Learning (ICML)*, 2025.
- **T. Truong***, N. Q. Pham*, Q. Tran, T. Nguyen, D. Phung, and T. Le. Promoting Ensemble Diversity with Interactive Bayesian Distributional Robustness for Fine-tuning Foundation Models. *International Conference on Machine Learning (ICML)*, 2025.
- H. Luo, **T. Truong**, T. Pham, M. Harandi, D. Phung, T. Le. Explicit Eigenvalue Regularization Improves Sharpness-Aware Minimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- C. Cameron, J. Hartford, T. Lundy, **T. Truong**, A. Milligan, R. Chen, K. Leyton-Brown. UNSAT Solver Synthesis via Monte Carlo Forest Search. In *International Conference on the Integration of Constraint Programming, Artificial Intelligence, and Operations Research (CPAIOR)*, pp. 170–189, 2024.
- H. Tran-The, S. Gupta, S. Rana, **T. Truong**, L. Tran-Thanh, S. Venkatesh. Expected Improvement for Contextual Bandits. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- H. Phan*, L. Tran*, Q. Tran, N. Tran, **T. Truong**, Q. Lei, N. Ho, D. Phung, T. Le. Beyond Losses Reweighting: Empowering Multi-Task Learning via the Generalization Perspective. In *International Conference on Computer Vision (ICCV)*, Highlight presentation, 2025.

Preprints

- **T. Truong***, N. T. Diep*, L. Q. Dung*, T. Dinh, H. Nguyen, N. Ho. *Hypernetwork-Driven Low-Rank Adaptation Across Attention Heads*. Under Review at *International Conference on Machine Learning (ICML)*, 2026.
- **T. Truong***, N. T. Diep*, H. Dang*, T. Dinh, H. Nguyen, N. Ho. *DoRAN: Stabilizing Weight-Decomposed Low-Rank Adaptation via Noise Injection and Auxiliary Networks*. Under Review at *International Conference on Machine Learning (ICML)*, 2026.

Research & Teaching Experience

Center for AI Research, VinUniversity
AI Research Resident

Hanoi, Vietnam
Dec 2025 – June 2026

Supervisor: Duy Nguyen (PhD Candidate at Max Planck Research School for Intelligent Systems & DFKI).

- Main research topic: Data-efficient vision-language-action (VLA) models for Robot Manipulation.

VinAI Research (now Qualcomm AI Research)
Research Resident

Hybrid
Feb 2023 – Jul 2025

Supervisor: Professors Trung Le (Monash University), Nhat Ho (UT Austin)

- Main research topic: developed robust and data-efficient Parameter-efficient Fine-tuning (PEFT) techniques for Large Foundation Models.
- Published four papers, accepted at ICML 2025 (3) and NeurIPS 2024 (1).
- Participated in an applied project with the SmartEdge team that builds the prompt-based open-set detection systems for VinHomes SmartCity apartments.

University of British Columbia
Research Assistant

Vancouver, Canada
Apr 2022 – Dec 2022

Supervisor: Professor Kevin Leyton-Brown

- Main research topic: developed boolean satisfiability solvers using reinforcement learning
- Co-authored the paper “UNSAT solver synthesis via Monte Carlo Forest Search” at CPAIOR 2024

Teaching Assistant

Jan 2023 – Apr 2023

- Teaching Assistant for the course CPSC 340: Machine Learning and Data Mining

FPT Software AI Center
Research Resident

Hybrid
May 2021 – Mar 2022

Supervisor: Professor Hung Tran-The

- Main research topic: Contextual bandits and Offline Reinforcement Learning.
- Co-authored the paper “Expected Improvements for Contextual Bandits,” published at NeurIPS 2022.

Grad Course Projects

CPSC533Y Visual Geometry with Deep Learning with Dr. Kwang Moo Yi

Winter 2023

- Independently authored a short conference-style paper, *Point Cloud Reconstruction with Hyperspherical Neural Points*, under the supervision of Dr. Kwang Moo Yi, proposing a novel point cloud reconstruction method that integrates hyperspherical embeddings into Neural Points to improve reconstruction quality.
- Demonstrated that the proposed Hyper Neural Points method outperforms Neural Points and PU-Net on the Sketchfab2 dataset across three quantitative metrics (Chamfer Distance, Hausdorff Distance, and point-to-surface error), while also producing qualitatively smoother and more visually coherent reconstructions.

Professional Services

Reviewer at ICML (2024-2026), ICLR (2024-2026), CVPR (2024-2025), NeurIPS (2026).